Optical Transport Failure Resolution - Huawei Access Node KR-ACC-02 (Warning Severity: ALARM_6284)

Node: KR-ACC-02 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_6284

Description

The Optical Transport Failure domain in Huawei Access nodes, such as KR-ACC-02, is a critical failure scenario that arises due to the degradation of SFP/XFP transceivers, fiber cuts, or loss of receive/transmit power. This failure is inherent to the nature of optical transport systems, which rely on the integrity of the fiber-optic infrastructure to transmit data between network segments. In the Huawei Access node, the optical transport failure is typically characterized by an alarm code of ALARM_6284, indicating a warning severity. This alarm is triggered when the node detects a degradation in the optical signal quality, which can be caused by various factors, including, but not limited to, SFP/XFP degradation, fiber cuts, or receive/transmit power loss. Understanding the underlying protocol and hardware mechanics of the optical transport system is crucial for diagnosing and resolving this type of failure.

Pre-requisite Checks

- Check: Verify the site location (KR Puram Access) and node ID (KR-ACC-02) for accuracy.
- Check: Ensure redundant link status is active and functioning properly.
- Check: Dispatch remote hands to the site location (KR Puram Access) for assistance, if required.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

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Command: display transceiver, Expected Output: Transceiver information, including status and optical power levels., Description: Run the display transceiver command to gather information about the SFP/XFP transceivers, including their status and optical power levels. This command helps to identify any potential issues with the transceivers, such as degradation or malfunction.

Command: display fiber-optic-connection, Expected Output: Fiber-optic connection information, including status and signal quality., Description: Run the display fiber-optic-connection command to gather
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information about the fiber-optic connections, including their status and signal quality. This command helps to identify any potential issues with the fiber-optic infrastructure, such as cuts or degradation.

Command: display optical-power, Expected Output: Optical power levels for receive and transmit ports., Description: Run the display optical-power command to gather information about the optical power levels for receive and transmit ports. This command helps to identify any potential issues with the optical signal quality, such as power loss or degradation.
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Resolution Steps

Step 1: Step: Replace SFP/XFP transceiver, Command: replace transceiver, Description: Replace the faulty SFP/XFP transceiver with a new one to resolve the issue. Ensure the replacement transceiver is compatible with the node and fiber-optic infrastructure.

Step 2: Step: Repair fiber-optic connection, Command: repair fiber-optic-connection, Description: Repair the faulty fiber-optic connection to resolve the issue. Ensure the repair is done by a qualified technician and the connection is properly tested before restoring service.

Step 3: Step: Adjust optical power levels, Command: adjust optical-power, Description: Adjust the optical power levels for receive and transmit ports to resolve the issue. Ensure the adjustment is done within the recommended range for the node and fiber-optic infrastructure.

Post-Check Verification

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Command: display transceiver, Expected Output: Transceiver information, including status and optical power levels., Description: Run the display transceiver command to verify the SFP/XFP transceivers are functioning properly and optical power levels are within the recommended range.

Command: display fiber-optic-connection, Expected Output: Fiber-optic connection information, including status and signal quality., Description: Run the display fiber-optic-connection command to verify the fiber-optic connections are functioning properly and signal quality is within the recommended range.

Command: display optical-power, Expected Output: Optical power levels for receive and transmit ports., Description: Run the display optical-power command to verify the optical power levels for receive and transmit ports are within the recommended range.
```

Roll-back Procedures

- Step: Restore original configuration, Command: undo replace transceiver, Description: Restore the original SFP/XFP transceiver to resolve the issue. Ensure the replacement transceiver is properly removed before restoring the original configuration.

- Step: Restore original fiber-optic connection, Command: undo repair fiber-optic-connection, Description: Restore the original fiber-optic connection to resolve the issue. Ensure the repair is properly undone before restoring the original connection.

- Step: Restore original optical power levels, Command: undo adjust optical-power, Description: Restore the original optical power levels for receive and transmit ports to resolve the issue. Ensure the adjustment is properly undone before restoring the original power levels.